

INCORPORATING FRAILTY EFFECTS IN THE COX PROPORTIONAL HAZARDS MODEL USING TWO INDEPENDENT METHODS IN TWO INDEPENDENT DATA SETS

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Summary: It is often of interest to compare the time that elapses before some event occurs. This type of analysis is known as survival analysis, and a well-known model that is used to account for covariates dependence, is the Cox proportional hazards model. However, there may be times when the Cox proportional hazards model does not adequately explain the distribution of survival times, and these deviations may sometimes be explained by unaccounted random heterogeneity, known as frailty. We consider the shared frailty model, which arises from situations where observations are correlated due to being exposed to common factors, such as coming from the same geographical area or the same hospital, for instance. We present two independent methods for fitting a one-parameter gamma frailty model: the penalized partial likelihood approach, and a Bayesian approach. These methods are then applied to two datasets and the results presented with discussion.

1. Introduction

In many statistical applications and analyses the variable of interest is often the time that elapses before some event occurs. This type of analysis is applicable in many areas in biological sciences such as animal and plant production in agriculture, observational and clinical trials in medical research, ecological studies, as well as in industrial processes to model failure times of items in production.

In survival analysis, the hazard function is used to model survival, or time-to-event data. This function evaluated at a given time gives the risk that an individual experiences an event, given survival to that time. A well-known model developed for dealing with survival time data is the Cox Proportional Hazards model (Cox, 1972). This model allows the incorporation of measured or observed covariates, such as age, gender, type of treatment and so on, to modify the hazard in order to capture observed heterogeneity. However, there may be times when the Cox Proportional Hazards model does not adequately describe the distribution of survival time. The deviations from the proportional hazards model may sometimes be explained by unaccounted random heterogeneity, otherwise known as frailty, first described by Vaupel *et al.* in 1979. A possible cause of this is when important covariates describing the survival of an individual are omitted (Keiding *et al.*, 1997). If standard methods are applied, the resulting estimates will be biased. To account for frailty in a model, an unmeasured random effect for each individual is incorporated in the hazard function, under the assumption that frailty is independent of any censoring that may take place. This is known as the univariate frailty model.

Practical situations arise where a cluster of observations are dependent or similar due to being associated with a common cluster such as a farm, clinic, repeated observations on an individual in longitudinal studies and so on. In the case where data are correlated due to a clustering effect, a frailty term is included in the model in order to account for the dependence in the data within a cluster. Ignoring the clustering effect and applying the standard Cox model can lead to biased estimates. We say that the observations from a given cluster, for example a farm, share a frailty specific to that farm, hence the term shared frailty model, also known as the multivariate frailty model. This paper will focus on the shared frailty model.

Another method of handling clustered data in the Cox model is that of using a robust sandwich covariance matrix estimate to account for the intracluster dependence (Lee *et al.*, 1992). This does not yield a change in the parameter estimates and thus the hazard ratios, however there may be changes in the estimates of the standard errors of the parameters, and accordingly may affect the statistical significance of the parameters.

2. The data

Two data sets were considered, both with clustering variables. The first data set is from the Fisher family history, a genealogy of the Old Order Amish Community based in Lancaster County, USA. This data has been previously published in King *et al.* (1996). The population was a closed and stable one, and had maintained extensive genealogical records. The data set consisted of 2860 individuals in 458 sibships, where individuals with the same sibship index are related. The percentage of censored individuals was 20.8% ($n = 594$). The outcome modelled was age at death. Gender and birth year were additional covariates. Gender was evenly distributed across the data, and birth year ranged

from 1801 to 1921. The outcome variable, age, ranged from 0 to 108 years at death. Sibship size ranged from 1 to 20, with the average sibship size being 6.24.

The second data set was obtained from a clinical trial in which a new treatment for lung cancer was compared to a standard treatment. The data was published in Gray (1995). The new treatment was denoted by ‘CAV-HEM’ and the standard by ‘CAV’. The clustering variable is the institution number, where the number of patients in each institution varied from 4 to 56, and the total number of institutions was 26. Other variables available for analysis were performance status (1 = performing well, 0 = not performing well), liver metastases (1 = yes, 0 = no), bone metastases (1 = yes, 0 = no) and weightloss (1 = yes, 0 = no). The variables were approximately evenly distributed over the treatment regimen. The outcome of interest was time to death.

3. The shared frailty model

The shared frailty model assumes that each individual within a cluster has the same frailty. To account for the correlation of failure times within a cluster, a random effect (or frailty effect) is included in the hazard function. The frailty model can be seen as a linear mixed effects model on a log hazard scale with a frailty term acting multiplicatively on the hazard function. The frailties are assumed to be independent between or across clusters, whilst the failure times of individuals within a cluster are dependent. However, conditional on frailties, the failure times are independent.

3.1 Model formulation

Suppose that there are n individuals, assigned to I groups, where the i^{th} group has n_i individuals such that $\sum_{i=1}^I n_i = n$. Suppose that the number of events

experienced by the i^{th} group is given by $D_i = \sum_{j=1}^{n_i} \delta_{ij}$, where δ_{ij} is the censoring indicator which takes on the value 1 when an event occurs and 0 when it does not. The hazard for the j^{th} individual from the i^{th} group is

$$h_{ij}(t) = h_0(t) \exp(x_{ij}^T \beta + w_i) \quad (1)$$

where x_{ij} is a vector of p covariates for individual j in group i , $h_0(t)$ is the baseline hazard, and w_i is the random effect for the i^{th} group. The w_i 's are an independently and identically distributed random sample from a density $f_W(\cdot)$. The model can be rewritten in the following manner:

$$\begin{aligned} h_{ij}(t) &= h_0(t) \exp(w_i) \exp(x_{ij}^T \beta) \\ &= z_i h_0(t) \exp(x_{ij}^T \beta) \end{aligned} \quad (2)$$

where $z_i = \exp(w_i)$ is known as the frailty term. The z_i 's are independent, and are assumed to have common density $f_Z(\cdot)$. The two commonly used distributions for the frailties are the log-normal density and the one-parameter gamma density for Z . In this paper, the one-parameter gamma density was used when fitting the model. The density is given by

$$f_Z(z) = \frac{\alpha^\alpha z^{\alpha-1} e^{-\alpha z}}{\Gamma(\alpha)} \quad (3)$$

with corresponding mean and variance of 1 and $\theta = 1/\alpha$ respectively. The variance, θ , is often the parameter of interest, and is the frailty parameter reported after the fitting of the model. The advantages to using the gamma distribution to model the frailty are that it is analytically tractable, readily computable and is one of the few distributions that are used to model variables that are positive. Since frailty cannot be negative the gamma distribution is thus a suitable candidate. Note that if w_i is omitted in equation (1) the standard Cox proportional hazards model is recovered.

3.2 Estimation in the frailty model

The two independent methods of estimation this paper will focus on are the penalized partial likelihood approach (Therneau and Grambsch, 2000) and the Bayes method of estimation using Markov Chain Monte Carlo (MCMC) (Vaida and Xu, 2000). Other methods of estimation in the literature include Expectation Maximisation (EM) algorithm (Klein, 1992), Monte Carlo EM (MCEM) approach (Ripatti *et al.*, 2002) and different methods using Laplace transformation (Ripatti and Palmgren, 2000, Cortinas Abrahantes and Burzykowski, 2004).

3.2.1 Penalized partial likelihood approach

Penalized partial likelihood estimation originates from cubic splines regression in the Cox proportional hazards model (Nguti, 2003). When using the penalized partial likelihood approach for estimation, the hazard function is formulated in terms of the random effects w_i rather than the frailties, z_i . The set of equations that govern the model for individual j and cluster i is given below.

$$h_{ij}(t) = h_0(t) \exp(x_{ij}^T \beta + w_i) \quad (4)$$

$$S_{ij}(t) = \exp[-\Lambda_0(t) \exp(x_{ij}^T \beta + w_i)] \quad (5)$$

$$L_i(w_i, t_{ij}; \alpha) = f_w(w_i) \times h_{ij}(t)^{\delta_{ij}} S_{ij}(t)$$

$$\ell_i(\beta, \alpha; w_i, t_{ij}) = \ln f_w(w_i) + \delta_{ij} \ln h_{ij}(t) + \ln S_{ij}(t) \quad (6)$$

where $h_{ij}(t)$ is the hazard function, $S_{ij}(t)$ is the survival function, $L_i(w_i, t_{ij}; \alpha)$ is the likelihood and $\ell_i(\alpha; w_i, t_{ij})$ the log-likelihood. Substituting equations (4) and (5) into the full data log-likelihood (equation

(6), gives the contribution for the j^{th} individual in the i^{th} cluster as

$$\begin{aligned}\ell(\alpha; w_i, t_{ij}) &= \ln f_w(w_i) + \delta_{ij} (\ln [h_0(t) \exp(x_{ij}^T \beta + w_i)]) \\ &\quad - \Lambda_0(t) \exp(x_{ij}^T \beta + w_i) \\ &= \ln f_w(w_i) + \delta_{ij} (\ln h_0(t) + x_{ij}^T \beta + w_i) \\ &\quad - \Lambda_0(t) \exp(x_{ij}^T \beta + w_i).\end{aligned}$$

The full data log-likelihood can be written as

$$\tilde{\ell}_{full}(\beta, \alpha, h_0) = \tilde{\ell}_{full,1}(\beta, h_0) + \tilde{\ell}_{full,2}(\alpha) \quad (7)$$

where

$$\begin{aligned}\tilde{\ell}_{full,1}(\beta, h_0) &= \sum_{i=1}^I \sum_{j=1}^J [\delta_{ij} (\ln h_0(t) + x_{ij}^T \beta + w_i) - \Lambda_0(t) \exp(x_{ij}^T \beta + w_i)] \\ \tilde{\ell}_{full,2}(\alpha) &= \sum_{i=1}^I \ln f_w(w_i)\end{aligned}$$

where $\ell_{full,2}(\alpha)$ can be seen as the penalty term, and the mean for the w_i 's is 0. For $w_i \ll 0$ or $w_i \gg 0$, $f_w(w_i)$ is small, and thus $\log f_w(w_i)$ takes a large negative value which in turn decreases the likelihood, in other words it acts like a penalty. Thus

$$\tilde{\ell}_{full,2}(\alpha) = -\ell_{pen}(\alpha) \quad (8)$$

with

$$\ell_{pen}(\alpha) = -\sum_{i=1}^I \ln f_w(w_i). \quad (9)$$

In order to apply semi-parametric ideas, consider the w_i 's in $\tilde{\ell}_{full,1}(\beta, h_0)$ as 'parameters' with corresponding covariates similar to that of a design matrix in the equation

$$Y = X\beta + Zw \quad (10)$$

where Z is the design matrix (Janssen, 2005). Using partial likelihood ideas, $\tilde{\ell}_{full,1}(\beta, h_0)$ is replaced by

$$\ell_{part}(\beta, w) = \sum_{i=1}^I \sum_{j=1}^J \delta_{ij} \left[\eta_{ij} - \ln \left(\sum_{k \in R(t_{ij})} \exp(\eta_k) \right) \right] \quad (11)$$

where

$$\eta_{ij} = x_{ij}^T \beta + w_i.$$

Thus in order to make inference on the parameters β and α , the following penalized partial likelihood is used

$$\ell_{ppl}(\beta, \alpha, w) = \ell_{part}(\beta, w) - \ell_{pen}(\alpha, w).$$

Application to loggamma density

If the frailties are assumed to be gamma distributed, then the random terms, the w_i 's, are loggamma distributed, with probability density function (p.d.f.)

$$f_w(w) = \frac{\alpha^\alpha (\exp(w))^\alpha \exp[-\alpha \exp(w)]}{\Gamma(\alpha)}$$

Taking the natural log of this p.d.f. results in

$$\ln f_w(w) = \alpha(w - \exp(w)) - (-\alpha \ln \alpha + \ln \Gamma(\alpha)).$$

Hence $\ell_{pen}(\alpha)$ is given by

$$\ell_{pen}(\alpha) = - \sum_{i=1}^I \{\alpha(w_i - \exp(w_i))\} + I(-\alpha \ln \alpha + \ln \Gamma(\alpha)).$$

The maximisation of the penalized partial likelihood consists of an inner and outer loop (Nguti, 2003). In the inner loop the rule is that given a provisional value for α , the Newton Raphson procedure is employed to maximise $\ell_{ppl}(\beta, \alpha, w)$ for β and w to obtain the best linear unbiased predictors. In the outer loop, a modified version of the log-likelihood is maximised for α . Let r denote the outer loop index, and k the inner loop index. Let $\alpha^{(r)}$ be the estimate for α at the r^{th} iteration of the outer loop. Then $\beta^{(r;k)}$

and $w^{(r;k)}$ are the estimates and predictions for β and w at the k^{th} iterative step, given $\alpha^{(r)}$. The starting value for β is obtained from the estimates from fitting a normal Cox model, and for starting values of $w^{(1;0)}$ and $\alpha^{(1)}$ the k^{th} iterative step is given by

$$\begin{bmatrix} \beta^{(r,k)} \\ w^{(r,k)} \end{bmatrix} = \begin{bmatrix} \beta^{(r,k-1)} \\ w^{(r,k-1)} \end{bmatrix} - V^{-1} \begin{bmatrix} 0 \\ [\alpha^{(r)}]^{-1} w^{(r,k-1)} \end{bmatrix} + V^{-1} [X \ Z] \frac{d\ell_{part}(\beta, w)}{d\eta}$$

where

$$\begin{aligned} V &= \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \\ &= \begin{bmatrix} X^T \\ Z^T \end{bmatrix} \left(\frac{-\partial^2 \ell_{part}(\beta, w)}{\partial \eta \partial \eta^T} \right) [X \ Z] + \begin{bmatrix} 0 & 0 \\ 0 & [\alpha^r]^{-1} I_I \end{bmatrix} \end{aligned}$$

where $X = [x_{11}, \dots, x_{In_I}]^T$ is an $n \times p$ covariate matrix with $n = \sum_{i=1}^I n_i$, $Z = \text{diag}(1_{n_1}, \dots, 1_{n_I})$ with 1_{n_i} as a column vector of size n_i with all entries one, and $\eta = X\beta + Zw$, such that $\eta^T = (\eta_{11}, \dots, \eta_{In_I})$ (Nguti, 2003).

Once the Newton Raphson procedure has converged for the current value of $\alpha^{(r)}$, the procedure moves to the outer loop of the algorithm. In the outer loop of the algorithm, a golden section search (Press *et al.*, 1992) is applied to a modified version of the log-likelihood in order to update the estimate of α . This likelihood is

$$\begin{aligned} \ell_{part,obs}(\beta, w) &= \ell_{part}(\beta, w) \\ &+ \sum_{i=1}^I \left[\ln \left(\frac{\Gamma(D_i + \alpha)}{\Gamma(\alpha)} \right) + \alpha \ln \left(\frac{\alpha}{\Lambda_i + \alpha} \right) \right. \\ &\quad \left. - D_i \ln(D_i + \alpha + D_i) \right] \end{aligned}$$

and the details of how this is derived are available in Therneau and Grambsch (2000).

The algorithm continues until the stopping criterion given by

$$\left| \ell_{part,obs} \left(\hat{\beta}^{(r)}, \alpha^{(r)}, \hat{w}^{(r)} \right) - \ell_{part,obs} \left(\hat{\beta}^{(r-1)}, \alpha^{(r-1)}, \hat{w}^{(r-1)} \right) \right| < \varepsilon^*$$

is reached.

3.2.2 The Bayesian approach

To fully specify the model from a Bayesian perspective, priors need to be assigned to the fixed effects, the random effect and the hyperparameter for the distribution of frailty. Following the procedure of Bolstad and Manda (2001), the prior for the fixed effects, β , is normally distributed with mean vector d_0 and diagonal covariance matrix $D_0 = cI$, where c is a number. The prior mean is assumed to be 0 because the fixed effects represent logarithms of relative risks (Bolstad and Manda, 2001). The frailty effect represents a relative risk, and thus should have a mean equal to 1. It is modelled as an independent draw from a gamma distribution with both parameters equal to α . Thus, given α , the mean and variance of the frailty effect is 1 and $\theta = 1/\alpha$. The hyperparameter is modelled as an independent draw from a gamma distribution that has parameters ν and κ (Bolstad and Manda, 2001).

The joint distribution of the data and the parameters is given by

$$\begin{aligned} f(data, \beta, z_i, \alpha) &= f(\beta) f(\alpha) \times \prod_{i=1}^I f(z_i | \alpha) \times \prod_{j=1}^J L_i(z_i, t_{ij}; \theta) \\ &= (2\pi)^{-p/2} |D_0|^{-\frac{1}{2}} \exp \left[-\frac{1}{2} (\beta - d_0)^T D_0^{-1} (\beta - d_0) \right] \\ &\quad \times \frac{\nu^\kappa}{\Gamma(\kappa)} \alpha^{\kappa-1} e^{-\nu\alpha} \\ &\quad \times \prod_{i=1}^I \frac{\alpha^\alpha}{\Gamma(\alpha)} z_i^{\alpha-1} e^{-\alpha z_i} \\ &\quad \times \prod_{j=1}^J \left(z_i h_0(t) e^{x_{ij}^T \beta} \right)^{\delta_{ij}} \exp \left(-z_i \Lambda_0(t) e^{x_{ij}^T \beta} \right). \end{aligned}$$

The analytic solution of the problem requires determining the posterior distribution of the parameters and hyperparameters, given the observed data. In other words, the parameters not of interest need to be integrated out of the joint distribution to obtain the marginals. For example, to obtain the marginal for β_1 , the following integration needs to be performed

$$f(\beta_1 | \text{data}, z_i, \alpha) = \int \dots \int f(\text{data}, \beta, z_i, \alpha) \partial\beta_2 \dots \partial\beta_k \partial z_i \partial\alpha.$$

This may not be possible to do analytically, and is impractical to do numerically, thus the approach is to find a Markov chain that has the posterior distribution as its long-run distribution (Bolstad and Manda, 2001). Thus a sample is drawn from the joint posterior density and the empirical version of the marginals of the posterior density are obtained. This is done using the Metropolis algorithm and Gibbs sampler. The estimates $\hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\alpha}$ are then taken as the medians of the empirical densities (Janssen, 2005).

Details of how the relevant Gibbs conditionals are directly computed from the joint distribution can be found in Zuma and Lurie (2005) and Bolstad and Manda (2001).

4. Application and discussion

Under this section the two independent methods (the PPL and Bayesian approaches) are applied to the two independent data sets. The software used for fitting the frailty models using the penalised partial likelihood approach was Stata 9.2, and WinBuGs14 was used for the Bayesian approach. It is noted that the two methods of incorporating frailty are conceptually different and cannot be directly compared.

4.1 Penalised partial likelihood approach

4.1.1 Old Order Amish data

The model fitted is

$$h_{ij}(t) = h_0(t) \exp(\beta_1 x_{sex} + \beta_2 x_{byr} + w_i)$$

where $i = 1, \dots, 458$, $j = 1, \dots, n_i$, $n = 2860$ and w_i is the random effect for sibship i . The model was fitted in Stata 9.2, with the final log-likelihood given as -13690.271 . The Efron approximation for handling ties was used. A shared gamma frailty for the clustering variable sibship was assumed. The frailty variance is given by θ . The parameter estimates are given in Table 1.

The frailty variance is 0.3262, which indicates that there is some variability in the frailty between the clusters. A likelihood ratio test of $\theta = 0$ has a χ^2 statistic of 172.09 on one degree of freedom, which has an associated p-value < 0.0001 . This is highly significant, indicating that the frailty effect should be included in the model. The other variables are also significant in the model at a 5% significance level. The interpretation of the sex variable is that the hazard of death for women is 0.8966 that of males, indicating that women live longer. The birth year variable has a coefficient of -0.0106, and hazard ratio of 0.9895. A useful statistic is $100(e^\beta - 1) = -1.05$. This means that for each one year increase in birth year, the hazard of death decreases by 1.05%.

Table 1 Parameter estimates for penalised partial likelihood approach

Variable	Parameter estimates	Std. error	z	p-value	Hazard ratio
sex	-0.1091	0.0492	-2.22	0.027	0.8966
byr	-0.0106	0.0017	-6.20	0.000001	0.9895
θ	0.3262	0.0434			

4.1.2 Lung cancer data

The model fitted is

$$h_{ij}(t) = h_0(t) \exp(\beta_1 x_{trt} + \beta_2 x_{perf\ stat} + \beta_3 x_{liver} + \beta_4 x_{bone} + \beta_5 x_{weightloss} + w_i)$$

where $i = 1, \dots, 26$, $j = 1, \dots, n_i$, $n = 570$ and w_i is the random effect for institution i . The model was fitted in Stata 9.2. The final log-likelihood was -2990.5 . The Efron approximation for handling ties was used, and a gamma shared frailty for the clustering variable sibship was assumed. The frailty variance is given by θ . The parameter estimates are given in Table 2. The frailty variance is very close to zero, implying that the clusters have a similar frailty, and thus not very heterogeneous. A likelihood ratio test of $\theta = 0$ has a χ^2 statistic $= 5.5 \times 10^{-10}$ on one degree of freedom, which has an associated p-value of 0.5. This is not significant and in this case including a random effect in the Cox model does not add any value to the model. The other variables in the model are significant at a 5% significance level. The new treatment (CAVHEM) has a reduced hazard compared to the old treatment (CAV); the hazard is approximately 77% that of the old treatment. The hazard for the patients with a good performance indicator is approximately 55% that of the patients who have a poor performance indicator. Patients who have liver metastases, bone metastases or weight loss have a higher hazard compared to the patients that do not.

Table 2 Parameter estimates for penalised partial likelihood approach

Variable	Parameter estimates	Std. error	z	p-value	Hazard ratio
trt	-0.257	0.0860	-2.99	0.0028	0.773
perfstat	-0.607	0.1044	-5.81	<0.00001	0.545
liver	0.419	0.0903	4.64	0.0000035	1.520
bone	0.233	0.0934	2.50	0.013	1.262
weightloss	0.203	0.0876	2.32	0.021	1.225
θ	1.21×10^{-18}	2.7×10^{-14}			

4.2 Bayesian approach

4.2.1 Old Order Amish data

The model is parameterised slightly differently to the PPL approach, using the frailties, z_i , instead of the random effects, w_i , given by the following equation:

$$h_{ij}(t) = h_0(t)z_i e^{x_{ij}^T \beta}.$$

The baseline hazard was modelled as $\text{Gamma}(\mu, c)$ and the frailty effect was modelled as $\text{Gamma}(\alpha, \alpha)$. The hyperparameter α was modelled as $\text{Gamma}(0.001, 0.001)$ and the fixed effects were modelled as $N(0, 0.000001)$. Three parallel chains were run and 20 000 iterations were computed. All the fixed effects parameters, a subset of the random effects, the variance of the random effects and baseline hazard were monitored for convergence. The first 1999 iterations were discarded for the burn-in. The Gelman-Rubin statistic, which was in a graphical format, was used to determine convergence, as well as the history plots that are produced in WinBuGs. The use of multiple chains allows one to monitor convergence of the parameters. Convergence appeared to be satisfied in this case, although this is a difficult and subjective task. Another measure of convergence is whether the ratio of the MC error and the standard deviation of the parameter estimates is less than 5%. The

estimates of the parameters were taken to be the median instead of the mean, as it is a more stable measure. Table 3 shows the results obtained. The frailty variance estimate is 0.5005 with a 95% confidence interval of (0.4338; 0.5708), indicating that there is variation between the clusters. The hazard ratio for gender is 0.883 and the interpretation is that the hazard for women is 88% that for the men. The Wald χ^2 statistic was used to test whether the parameters were significant, and it was found that both sex and birth year were significant. It is also important to note that the ratio of the MC error and the standard deviation is less than 5% for each variable; which implies convergence has been reached.

Table 3 Bayesian parameter estimates for family data

Parameter	Mean	SD	M C Error	Median	MCE/SD $\times$ 100
sex	-0.1239	0.04555	0.0008363	-0.124	1.836%
byr	-0.002364	0.0005855	0.00002905	-0.002474	4.962%
θ	0.501	0.03519	0.0005216	0.5005	1.48%

4.2.2 Lung cancer data

The baseline hazard was modelled as $\text{Gamma}(\mu, c)$, the frailty was modelled as $\text{Gamma}(\alpha, \alpha)$, the hyperparameter α was modelled as $\text{Gamma}(0.001, 0.001)$ and the fixed effects were modelled as $N(0, 0.000001)$. Three parallel chains were run and 5000 iterations were computed. All the fixed effects parameters, a subset of the random effects, the variance of the random effects and baseline hazard were monitored for convergence. The burn-in was taken to be the first 1000 iterations. The model converged relatively quickly compared to the family data model. Table 4 contains all the estimates of the parameters. Using the Wald test the frailty variance θ is not statistically significant, indicating that there is not much variation between the institutions. All the monitors for the parameter estimates indicated that convergence had been satisfied. The ratio of the MC error and the standard deviation was less than 5% for all parameter

estimates. Using the Wald χ^2 statistic to test for significance of parameters it was found that the variables are all significant.

Table 4 Bayesian parameter estimates for lung cancer data

Parameter	Mean	SD	M C Error	Median	MCE/SD $\times$ 100
trt	-0.248	0.08653	0.001387	-0.2477	1.603%
perfstat	-0.6023	0.1056	0.002777	-0.6008	2.630%
liver	0.4182	0.09103	0.001208	0.4189	1.327%
bone	0.2339	0.09314	0.00114	0.2342	1.224%
weightloss	0.2067	0.08934	0.001646	0.2053	1.842%
θ	0.07179	0.04094	0.001785	0.06162	4.36%

4.3 Discussion of the different methods

4.3.1 Old Order Amish data

The models fitted to the data were the Cox proportional hazards (Cox PH) model without random effects (Phipson, 2006), the frailty model estimated using the penalised partial likelihood approach and the frailty model estimated using Bayesian methods. From Table 5 it can be seen that generally the magnitude and sign of the parameter estimates of gender are quite similar. The standard Cox model and the Cox model with frailty have very similar estimates for gender, with the Bayes method giving a slightly larger estimate. This could be due to the fact that the former two methods are both frequentist approaches and have a similar method of estimation. The standard errors in all three methods are quite similar despite the difference in approach between the first two methods and the third. The two parameter estimates of birth year under the standard Cox model and the PPL approach are quite similar as expected. However, the corresponding estimate of birth year under the Bayesian approach is quite different, a result that should not be a surprise because of the fundamental difference in the methods. In the standard Cox

model each later birth year has a 0.7% reduction in the hazard of death, which is closer to 1.05% for the PPL approach than 0.2% for the Bayesian approach. An alternative interpretation is that the percentage change in hazard of death between two birth years 10 years apart is a 10.5% reduction under the PPL approach compared to 7% under the Cox PH. The corresponding estimate is a 2% change under the Bayesian approach. In all three methods the Wald test shows that the parameter estimates of both sex and birth year are statistically significant at the 5% level. What is noticeably different in the results is the estimate of frailty variance (θ). It is larger using the Bayesian method than the PPL method. Again as stated earlier the two results are not directly comparable because the Bayesian approach incorporates a prior to the unknown variance parameter while the PPL does not. However, this bias may reflect the incapability of the PPL approach to take into consideration the uncertainty due to the estimation of parameters describing random effect distributions (Ripatti and Palmgren, 2000) in general. Nonetheless both estimation methods show that the survival experience of the participants vary considerably across families.

Table 5 Parameter estimates for the different approaches

Parameter	Cox PH		PPL		Bayes	
	Estimate	SD	Median	SD	Median	SD
sex	-0.11186	0.042185	-0.1091	0.0492	-0.1239	0.04555
byr	-0.00698	0.000877	-0.0106	0.0017	-0.002364	0.0005855
θ	—	—	0.3262	0.0434	0.5005	0.03519

4.3.2 Lung cancer data

The estimates of the fixed effects parameters and standard errors are highly similar in all three models. The results are shown in Table 6. The main difference is in the estimation of the frailty effect variance under the two

independent approaches namely the PPL model and the Bayesian model. In the PPL model it is much smaller in magnitude than in the Bayesian model. However, as already concluded under the individual applications, the test of significance showed that the frailty variance is not significantly different from zero. In this case the Cox proportional hazards model without frailty is adequate for this data set.

Table 6 Parameter estimates for the different approaches

Parameter	Cox PH		PPL		Bayes	
	Estimate	SD	Median	SD	Median	SD
trt	-0.257	0.0860	-0.257	0.0860	-0.2477	0.08653
perfstat	-0.607	0.1044	-0.607	0.1044	-0.6008	0.1056
liver	0.419	0.0903	0.419	0.0903	0.4189	0.09103
bone	0.233	0.0934	0.233	0.0934	0.2343	0.09314
weightloss	0.203	0.0876	0.203	0.0876	0.2053	0.08934
θ	—	—	1.21×10^{-18}	2.73×10^{-14}	0.07179	0.04094

5. Conclusion

This paper has compared the Cox proportional hazards model with and without frailty effects using two independent data sets. The frailty effect was incorporated using two independent estimation methods namely the PPL and the Bayesian approaches. By not incorporating frailty effects in the model to account for correlation in the data, there is a risk that the standard errors of the fixed effects are under-estimated and this may lead to misleading results and conclusions. Further it was seen that caution must be upheld in relating parameter estimates for fixed effects from the two independent methods of incorporating frailty, namely the frequentist and Bayesian approaches. The two are fundamentally and conceptually different and should not be directly compared. From a computational perspective the penalized partial likelihood approach to estimating frailty models is readily available in many software

packages including R and Stata, whereas the Bayesian method requires more user input and programming via still readily available platforms such as WinBUGs and add-on packages in R. The latter is far more computationally intensive and the issue of whether convergence has been reached is an important one, as estimates obtained from a model which has not fully converged will be incorrect. Finally, it is hoped that the work will provide a useful reference to students and researchers in the area of modelling time-to-event data.

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